Note for derived SymTFT and chromatic homotopy

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1 Algebraic geometry

1.1 Derived self-intersections

1.1.1 Derived intersection of points

Goal

Compute the derived intersection of two points $\mathbf{0}, \boldsymbol{a} \in \mathbb{A}^n$ from a Koszul resolution. It is empty for $\boldsymbol{a} \neq \mathbf{0}$; for $\boldsymbol{a} = \mathbf{0}$ it is $\mathrm{Tor}_\bullet = \bigwedge^\bullet k^n$, the exterior algebra on n generators of degree 1, i.e. functions on the shifted tangent space $T_{\mathbf{0}}\mathbb{A}^n[-1]$.

Here, we compute the derived intersection of two points on $\mathbb{A}^n = \operatorname{Spec} k[x_1, x_2, \ldots, x_n]$. Let one point lie at $\mathbf{0} \in \mathbb{A}^n$, and the other at $\boldsymbol{a} \in \mathbb{A}^n$. The function algebra of the intersection is given by the derived tensor product

$$k_{(\mathbf{0})} \otimes^{\mathbb{L}}_{k[x_1,x_2,\ldots,x_n]} k_{(\boldsymbol{a})} \tag{1}$$

where $k_{(\boldsymbol{a})}$ is a $k[x_1, x_2, \ldots, x_n]$-module on which x_i acts as a_i. Denoting $k[x_1, x_2, \ldots, x_n] = R$ and $x_i - a_i = f_i$, we use the free R-modules $V = R^n$ and $\bigwedge^\bullet V$ to form a Koszul resolution of $k_{(\boldsymbol{a})}$. Let $\{e_1, e_2, \ldots, e_n\}$ be a chosen R-basis of V.

$$\begin{array}{ccccccccccc} \bigwedge^n V & \longrightarrow & \cdots & \longrightarrow & \bigwedge^3 V & \xrightarrow{\iota_f} & \bigwedge^2 V & \xrightarrow{\iota_f} & V & \xrightarrow{\iota_f} & R \\ & & & & & & & & & & \downarrow \\ & & & & & & & & & & k_{(\boldsymbol{a})} \end{array} \tag{2}$$

where the differential $d = \iota_f$ acts on the canonical basis as

$$d(e_{i_1} \wedge \cdots \wedge e_{i_p}) = \sum_{j=1}^{p} (-1)^{j-1} f_{i_j} e_{i_1} \wedge \cdots \widehat{e_{i_j}} \cdots \wedge e_{i_p}. \tag{3}$$

Tensoring with $k_{(\mathbf{0})} = k$, we get

$$k^{\binom{n}{n}} \longrightarrow \cdots \longrightarrow k^{\binom{n}{3}} \xrightarrow{\iota_{-a}} k^{\binom{n}{2}} \xrightarrow{\iota_{-a}} k^n \xrightarrow{\iota_{-a}} k \tag{4}$$

For $\boldsymbol{a} \neq \mathbf{0}$ the homology vanishes, indicating that the intersection is empty. For $\boldsymbol{a} = \mathbf{0}$ the morphisms are trivial, and the complex itself gives the homology. The algebra structure is inherited from that of Koszul complex, which yields the exterior algebra $\bigwedge^\bullet k^n$.

1.1.2 Self-intersection of embedding

Goal

Replace the point by a regular embedding $Z \subset X$ with ideal $I = (f_1, \dots, f_c)$ and show $\operatorname{Tor}_i^{\mathcal{O}_X}(\mathcal{O}_Z, \mathcal{O}_Z) = \bigwedge^i N_{Z/X}^\vee$, i.e. $Z \times_X^h Z = N_{Z/X}[-1]$. The only non-formal input is $(A/I)^c \simeq I/I^2$, which uses the regularity of $(f_1, \dots, f_c)$ (paragraph *Regularity and flatness*).

We begin by generalizing the result of §1.1.1 by replacing the point by a regular embedding $Z \subset X$. Locally, the embedding corresponds to some morphism of algebras $A \to A/I$ with $I = (f_1, \dots, f_c)$. Using the free A-module $V = A^c$, one obtains a Koszul resolution analogous to Eq. (3).

Remark 1.1. *The claim here is that Eq. (10b) below is a free resolution of A/I. See Lemma 1.2.*

Tensoring with A/I, the differentials are reduced to 0 so that the chain complex directly gives $\operatorname{Tor}_\bullet^A(A/I, A/I)$:

$$(A/I)^{\binom{c}{c}} \longrightarrow \cdots \longrightarrow (A/I)^{\binom{c}{2}} \xrightarrow{0} (A/I)^c \xrightarrow{0} A/I \tag{5}$$

Since the generators of $(A/I)^c$ over A/I correspond to the elements f_i, $(A/I)^c$ can be identified with I/I^2 (see below). Thus,

$$\operatorname{Tor}_\bullet^A(A/I, A/I) = \bigwedge^\bullet (I/I^2). \tag{6}$$

Sheafifying the above construction, we get

$$\mathcal{O}_Z \otimes^{\mathbb{L}}_{\mathcal{O}_X} \mathcal{O}_Z = \bigwedge^\bullet N_{Z/X}^\vee = \operatorname{Sym}(N_{Z/X}^\vee[1]) \quad \Longrightarrow \quad Z \times_X^h Z = N_{Z/X}[-1]. \tag{7}$$

Regularity and flatness. Why $(A/I)^c \simeq I/I^2$ for a regular sequence $I = (f_1, \dots f_c)$? Let A be a k-algebra and let $S := k[X_1, \dots, X_c]$ act on A by $X_i \mapsto f_i$. Let $J = (X_1, \dots, X_c) < S$. Then $J/J^2 = \bigoplus_{i=1}^c (S/J)\bar{X}_i \simeq (S/J)^c$ and hence $(J/J^2) \otimes_S A \simeq (S/J)^c \otimes_S A \simeq (A/I)^c$. Tensoring the short exact sequence (SES) $J^2 \to J \to J/J^2$ with A, one gets an exact sequence

$$J^2 \otimes_S A \longrightarrow J \otimes_S A \longrightarrow (J/J^2) \otimes_S A \longrightarrow 0 \tag{8}$$

It suffices to show that $J \otimes_S A \simeq I$ canonically, in such a way that the image of $J^2 \otimes_S A$ is I^2. To this end, tensor the SES $J \to S \to S/J$ with A:

$$\begin{array}{ccccccccc} \operatorname{Tor}_1^S(S/J, A) & \longrightarrow & J \otimes_S A & \longrightarrow & S \otimes_S A & \longrightarrow & (S/J) \otimes_S A & \longrightarrow & 0 \\ & & & & \| & & \| & & \\ & & & & A & \longrightarrow & A/I & & \end{array} \tag{9}$$

$\operatorname{Tor}_1^S(S/J, A) = 0$ will imply the desired result. The Koszul resolution of $S/J = k$ is familiar from §1.1.1:

$$\bigwedge^c S^c \longrightarrow \cdots \longrightarrow \bigwedge^3 S^c \xrightarrow{\iota_X} \bigwedge^2 S^c \xrightarrow{\iota_X} S^c \xrightarrow{\iota_X} S \tag{10a}$$

$$\overset{-\otimes_S A}{\rightsquigarrow} \quad \bigwedge^c A^c \longrightarrow \cdots \longrightarrow \bigwedge^3 A^c \xrightarrow{\iota_f} \bigwedge^2 A^c \xrightarrow{\iota_f} A^c \xrightarrow{\iota_f} A \tag{10b}$$

For $c = 1$, the first homology is nothing but $\ker(f\cdot)$, which vanishes if and only if f is a nonzerodivisor (i.e. (f) is regular). For $c > 1$, we proceed by induction. Denote the Koszul complex in Eq. (10b) by $K = C(f_1, \dots, f_c)$ and let $K' = C(f_1, \dots, f_{c-1})$. Then $K = K' \oplus K' \wedge e_c$. From the SES of A-modules $K' \to K \to K'[1]$ one obtains the long exact sequence of homology

$$\cdots \longrightarrow H_p(K') \xrightarrow{\pm f_c \cdot} H_p(K') \longrightarrow H_p(K) \longrightarrow H_{p-1}(K') \xrightarrow{\pm f_c \cdot} H_{p-1}(K') \longrightarrow \cdots \tag{11}$$

where the connecting homomorphisms $\delta = \pm f_c \cdot$ can be derived from

$$\mathrm{d}y = 0 \quad \Longrightarrow \quad \mathrm{d}(y \wedge e_c) = \pm y\,(\mathrm{d}e_c) = \pm f_c\, y. \tag{12}$$

By induction hypothesis we may assume that $H_1(K') = 0$. Then Eq. (11) states that $H_1(K) = \mathrm{Tor}_1^S(S/J, A) = 0$ if and only if f_c is a nonzerodivisor in $H_0(K') = A/(f_1, \dots, f_{c-1})$, which is precisely the regularity of $(f_1 \dots, f_c)$. In fact, the proof shows that $\mathrm{Tor}_{\geq 1}^S(S/J, A) = 0$, which is needed for Remark 1.1. More precisely, we interpret Eq. (10) in terms of the following lemma.

Lemma 1.2 (Tor-independent base change)**.** *Let $P_\bullet \to N$ be a free resolution of an S-module N, and let A be an S-algebra such that $\mathrm{Tor}_p^S(N, A) = 0$ for all $p \geq 1$. Then $P_\bullet \otimes_S A \to N \otimes_S A$ is a free resolution of the A-module $N \otimes_S A$.*

Proof. Each $P_p \otimes_S A$ is a free A-module. By the definition of Tor, $H_p(P_\bullet \otimes_S A) = \mathrm{Tor}_p^S(N, A)$, which vanishes for $p \geq 1$, while $H_0(P_\bullet \otimes_S A) = N \otimes_S A$ by right exactness of $- \otimes_S A$. Exactness is a property of the underlying complex of abelian groups, so it does not matter whether $P_\bullet \otimes_S A$ is regarded as a complex of S-modules or of A-modules. □

Applying Lemma 1.2 to the Koszul resolution of $N = S/J$, for which $N \otimes_S A = A/I$, shows that Eq. (10b) is a free resolution of A/I, as required in Remark 1.1.

1.1.3 Serre intersection formula (1)

Goal

For a curve D on a smooth surface S, make $\mathrm{Tor}_1^{\mathcal{O}_S}(\mathcal{O}_D, \mathcal{O}_D) = N_{D/S}^\vee = \mathcal{O}_D(-D)$ explicit through transition functions, and show that the Euler characteristic of the derived self-intersection recovers the classical number without moving D: $\chi(\mathcal{O}_D \otimes^{\mathbb{L}} \mathcal{O}_D) = \deg N_{D/S} = D \cdot D$. Examples: a line in $\mathbb{P}^2$ $(+1)$ and the exceptional curve of $\mathrm{Bl}_p \mathbb{P}^2$ (-1), which cannot be moved.

In §1.1.2 we identified the derived self-intersection of a regular embedding at the level of sheaves, $\mathrm{Tor}_i^{\mathcal{O}_X}(\mathcal{O}_Z, \mathcal{O}_Z) = \bigwedge^i N_{Z/X}^\vee$. This problem extracts a number from that identification: for a curve D on a surface S, the classical self-intersection number $D \cdot D$, which is computed by moving D to a linearly equivalent curve, is already encoded in the derived self-intersection and can be read off by taking the Euler characteristic, without moving anything. We first specialize Eq. (7) to $c = 1$ and make Tor_1 explicit as a line bundle on D. Since D has codimension 1, the derived tensor product contains Tor_1 and no more, and Eq. (7) reads

$\operatorname{Tor}_1^{\mathcal{O}_S}(\mathcal{O}_D, \mathcal{O}_D) = N_{D/S}^\vee = \mathcal{I}/\mathcal{I}^2$. Because S is smooth, D is locally principal: on a small open set U, $\mathcal{I}|_U = (f)$ for a nonzerodivisor f, so that

$$\mathcal{O}_U \xrightarrow[\sim]{f\cdot} \mathcal{I}|_U. \tag{13}$$

The local equation is unique only up to a unit, $f' = uf$ with $u \in \mathcal{O}^\times(U)$, and on an overlap $U_i \cap U_j$ the local equations are related by $f_i = g_{ij} f_j$. Hence $\mathcal{I}$ is a line bundle with transition functions g_{ij}, denoted $\mathcal{O}_S(-D)$; in general it has no global trivialization, since a global equation f would force $N_{D/S}^\vee \simeq \mathcal{O}_D$. Restricting to D,

$$N_{D/S}^\vee = \mathcal{I}/\mathcal{I}^2 = \mathcal{I} \otimes_{\mathcal{O}_S} \mathcal{O}_S/\mathcal{I} = \mathcal{I} \otimes_{\mathcal{O}_S} \mathcal{O}_D = \mathcal{O}_D(-D), \tag{14}$$

a line bundle on D with transition functions $g_{ij}|_D$.

Example 1.3 (line in $\mathbb{P}^2$)**.** *Consider $D = \ell = \{x_2 = 0\} = \mathbb{P}^1 \subset \mathbb{P}^2$. The ideal sheaf is generated by f_i on $U_i = \{x_i \neq 0\}$ $(i = 0, 1)$, where $f_0 = x_2/x_0$, $f_1 = x_2/x_1$. The transition function $g_{01} = x_1/x_0$ specifies the sheaf $\mathcal{O}_D(-D) = \mathcal{O}_{\mathbb{P}^1}(-1)$.*

Example 1.4 (exceptional line)**.** *Consider the exceptional line $E \subset \operatorname{Bl}_p \mathbb{P}^2$. A neighborhood of p is isomorphic to $\operatorname{Bl}_0 \mathbb{A}^2 = \{((x, y), [a : b]) \in \mathbb{A}^2 \times \mathbb{P}^1 \,:\, xb = ya\}$ with $\pi((x, y), [a : b]) = (x, y)$. Identify $E = \pi^{-1}(0, 0) = \mathbb{P}^1$. On $U_a = \{a \neq 0\} \subset E$, the conormal sheaf is spanned by x since $y = x(b/a)$ on $\{a \neq 0\} \subset \operatorname{Bl}_0 \mathbb{A}^2$. On U_b, the same sheaf is spanned by y as $x = y(a/b)$. The transition function $g_{ab} = x/y = a/b$ specifies the sheaf $\mathcal{O}_E(-E) = \mathcal{O}_{\mathbb{P}^1}(1)$.*

To get $\ell \cdot \ell$ in Example 1.3, take $\ell' = \{x_1 = 0\} \sim \ell$. See $x_1|_\ell$ as a section of $\mathcal{O}_\ell(1) = \mathcal{O}_{\mathbb{P}^2}(\ell)|_\ell = N_\ell$, whose zero locus is $\ell \cap \ell'$. While it has no zeros on U_1, the expression $x_1 = x_0(x_1/x_0)$ indicates that it has a single zero on U_0. Thus $\ell \cdot \ell = \ell \cdot \ell' = 1$.

However, this classical method fails for Example 1.4: E cannot be moved, since $H^0(E, N_E) = H^0(\mathbb{P}^1, \mathcal{O}(-1)) = 0$, and the sequence $0 \to \mathcal{O}_S \to \mathcal{O}_S(E) \to N_E \to 0$ then gives $H^0(S, \mathcal{O}_S(E)) = H^0(S, \mathcal{O}_S) = k$ (S is connected and projective).

The derived self-intersection needs no motion. Its homology sheaves $\operatorname{Tor}_0 = \mathcal{O}_D$ and $\operatorname{Tor}_1 = N_{D/S}^\vee$ are sheaves on D, and taking the alternating sum of their Euler characteristics, with Riemann–Roch on D in the form $\chi(L) - \chi(\mathcal{O}_D) = \deg L$ [5, Thm. IV.1.3], gives

$$\chi(\mathcal{O}_D \otimes^{\mathbb{L}} \mathcal{O}_D) = \chi(\mathcal{O}_D) - \chi(\mathcal{O}_D(-D)) = \deg \mathcal{O}_S(D)|_D = D \cdot D, \tag{15}$$

where the last equality is the standard formula for the intersection pairing on a surface [5, Ch. V, §1]. The reason is visible from a moved copy $D' = \operatorname{div}(s') \sim D$ meeting D properly: the sequence $0 \to \mathcal{O}_D(-D) \xrightarrow{s'|_D} \mathcal{O}_D \to \mathcal{O}_{D \cap D'} \to 0$ shows that the honest intersection $D \cap D'$ has the same alternating sum as the derived self-intersection. For the two examples, $\ell \cdot \ell = -\deg \mathcal{O}_{\mathbb{P}^1}(-1) = 1$ and $E \cdot E = -\deg \mathcal{O}_{\mathbb{P}^1}(1) = -1$, the latter in agreement with the classical value obtained only through auxiliary divisors, $E \sim \pi^* L' - \tilde{L}$.

1.1.4 Serre intersection formula (2)

Goal

For $Z = P_1 \cup P_2$, two planes meeting at the origin of $\mathbb{A}^4$, and a third plane W, show that $\mathrm{Tor}_0(\mathcal{O}_Z, \mathcal{O}_W)$ alone overcounts the intersection multiplicity (len = 3), while Serre's alternating sum $\sum_i (-1)^i \operatorname{len} \mathrm{Tor}_i(\mathcal{O}_Z, \mathcal{O}_W) = 3 - 1 = 2$ gives the right answer. Tools: the Mayer–Vietoris sequence of Z and Koszul resolutions.

In this section, we compute the intersection between two surfaces Z, W in $\mathbb{A}^4$. Let $R = k[x, y, z, w]$, $P_1 = V(x, y)$, $P_2 = V(z, w)$, $Z = P_1 \cup P_2$ and $W = V(x - z, y - w)$. Since $Z \cap W = 0$ and $\dim Z + \dim W = 4$, the intersection is proper. The expected multiplicity is 2 since W intersects with each of P_1, P_2 transversally.

$I(Z) = I(P_1)I(P_2) = (xz, xw, yz, yw)$. Naively, the function algebra of the intersection is expected to be

$$\mathcal{O}_Z \otimes \mathcal{O}_W = k[x, y, z, w]/(xz, xw, yz, yw, x - z, y - w) = k[x, y]/(x^2, xy, y^2) \tag{16}$$

which is 3-dimensional over k, not matching the expected multiplicity 2. Serre's formula repairs the discrepancy through higher torsion groups. To this end, we employ the Mayer–Vietoris sequence

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathcal{O}_Z & \longrightarrow & \mathcal{O}_{P_1} \oplus \mathcal{O}_{P_2} & \longrightarrow & \mathcal{O}_{P_1 \cap P_2} & \longrightarrow & 0 \\ & & & & & & \| & & \\ & & & & & & k & & \end{array} \tag{17}$$

and the corresponding long exact sequence of torsion groups

$$\begin{array}{ccccccc} \cdots \longrightarrow & \mathrm{Tor}_2(\mathcal{O}_Z, \mathcal{O}_W) & \longrightarrow & \mathrm{Tor}_2(\mathcal{O}_{P_1} \oplus \mathcal{O}_{P_2}, \mathcal{O}_W) & \longrightarrow & \mathrm{Tor}_2(k, \mathcal{O}_W) & \\ & \mathrm{Tor}_1(\mathcal{O}_Z, \mathcal{O}_W) & \longrightarrow & \mathrm{Tor}_1(\mathcal{O}_{P_1} \oplus \mathcal{O}_{P_2}, \mathcal{O}_W) & \longrightarrow & \mathrm{Tor}_1(k, \mathcal{O}_W) & \\ & \mathrm{Tor}_0(\mathcal{O}_Z, \mathcal{O}_W) & \longrightarrow & \mathrm{Tor}_0(\mathcal{O}_{P_1} \oplus \mathcal{O}_{P_2}, \mathcal{O}_W) & \longrightarrow & \mathrm{Tor}_0(k, \mathcal{O}_W) & \longrightarrow 0 \end{array} \tag{18}$$

We compute the second and the third columns as an input. Begin with a Koszul resolution of $\mathcal{O}_W$

$$\begin{array}{ccccccc} 0 & \longrightarrow & \bigwedge^2 R^2 & \longrightarrow & R^2 & \longrightarrow & R \\ & & & & & & \downarrow \\ & & & & & & R/(x - z, y - w) \end{array} \tag{19}$$

where the horizontal maps are $\iota_{(x-z,y-w)}$ as defined in Eq. (3). Tensoring with k, read the sequence of $\mathrm{Tor}_i(k, \mathcal{O}_W)$:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & k & \xrightarrow{0} & k^2 & \xrightarrow{0} & k & & \\ & & \| & & \| & & \| & & \\ & & \mathrm{Tor}_2 & & \mathrm{Tor}_1 & & \mathrm{Tor}_0 & & \end{array} \tag{20}$$

Tensoring (19) with $\mathcal{O}_{P_j}$ (WLOG $j = 2$), obtain the sequence of $\mathrm{Tor}_i(\mathcal{O}_Z, \mathcal{O}_W)$:

$$\begin{array}{ccccccc} 0 & \longrightarrow & k[x,y]e_1 \wedge e_2 & \xrightarrow{\iota_{(x,y)}} & k[x,y]e_1 \oplus k[x,y]e_2 & \xrightarrow{\iota_{(x,y)}} & k[x,y] \\ & & \mathrm{Tor}_2 = 0 & & \mathrm{Tor}_1 = 0 & & \mathrm{Tor}_0 = k \end{array} \tag{21}$$

Hence, (18) reads

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathrm{Tor}_2(\mathcal{O}_Z, \mathcal{O}_W) & \longrightarrow & 0 & \longrightarrow & k \\ & & \mathrm{Tor}_1(\mathcal{O}_Z, \mathcal{O}_W) & \longrightarrow & 0 & \longrightarrow & k^2 \\ & & \mathrm{Tor}_0(\mathcal{O}_Z, \mathcal{O}_W) & \longrightarrow & k^2 & \xrightarrow{\Delta} & k \longrightarrow 0 \end{array} \tag{22}$$

where $\Delta(a, b) = a - b$. It follows that $\mathrm{Tor}_2 = 0$, $\mathrm{Tor}_1 \simeq k$, $\mathrm{Tor}_0 \simeq k^3$. This confirms the Serre's formula

$$Z \cdot W = \sum_i (-1)^i \dim_k \mathrm{Tor}_i = 3 - 1 = 2. \tag{23}$$

1.1.5 Cotangent complex

Goal

See what the cotangent complex $\mathbb{L}$ adds to Ω^1. For a point of $\mathbb{A}^n$, $\mathbb{L}_{k/R} = k^n[1]$, so §1.1.1 is $\mathrm{Sym}\,\mathbb{L}_{k/R}$. For the fat point $k[x]/(x^2)$, $H_1(\mathbb{L}) \neq 0$ detects the singularity and gives the first-order deformation $T^1 = k$ that Ω^1 misses. For the non-lci $k[x, y]/(x^2, xy)$, $\mathbb{L}$ is no longer a two-term complex.

Cotangent complex of pt $\hookrightarrow \mathbb{A}^n$. The Koszul resolution of k over $R = k[x_1, \ldots, x_n]$ is

$$K = \left(\begin{array}{ccccccc} \bigwedge^n V & \xrightarrow{\iota_x} & \cdots & \xrightarrow{\iota_x} & \bigwedge^2 V & \xrightarrow{\iota_x} & V \xrightarrow{\iota_x} R \\ & & & & & & \| \\ & & & & & & \bigoplus_{i=1}^n R\varepsilon_i \end{array} \right) \xrightarrow{\sim} k, \tag{24a}$$

$$\rightsquigarrow \quad \Omega^1_{K/R} = \left(\begin{array}{ccccccc} \bigwedge^n V \otimes W & \xrightarrow{\iota_x} & \cdots & \xrightarrow{\iota_x} & V \otimes W & \xrightarrow{\iota_x} & W \longrightarrow 0 \\ & & & & & & \| \\ & & & & & & \bigoplus_{i=1}^n R\,\mathrm{d}\varepsilon_i \end{array} \right) = \bigoplus_{i=1}^n K\,\mathrm{d}\varepsilon_i \tag{24b}$$

$$\overset{-\otimes_K k}{\rightsquigarrow} \quad \mathbb{L}_{k/R} = \bigoplus_{i=1}^n k\,\mathrm{d}\varepsilon_i \ \ (\text{degree } 1) \ = \ k^n[1]. \tag{24c}$$

Here $\mathrm{d} : K \to \Omega^1_K$ has degree 0, $|\mathrm{d}x| = |x|$, and the differential of Ω^1_K is determined by $d\,\mathrm{d} = \mathrm{d}\,d$ together with the Leibniz rule ($d = \iota_x$). Then $d\varepsilon_i = x_i$ and $\mathrm{d}(d\varepsilon_i) = \mathrm{d}x_i = 0$ since $x_i \in R$, so the differential of $\Omega^1_{K/R} = \bigoplus_i K\,\mathrm{d}\varepsilon_i$ is $\iota_x \otimes 1$, that of K on each summand. Since $K \to k$ kills every ε_i, $\bigwedge^p V \otimes_K k = 0$ for $p \geq 1$, and only the degree-1 generators $\mathrm{d}\varepsilon_i$ survive.

Note 1.5. *The answer to §1.1.1 is* $\bigwedge^\bullet k^n = \mathrm{Sym}(k^n[1]) = \mathrm{Sym}(\mathbb{L}_{k/R})$.

Cotangent complex of lci. To generalize to regular embeddings $Z \subset X$, one starts with the resolution (10b). With a regular sequence $f = (f_1, \dots, f_c)$,

$$K = \left(\begin{array}{ccccccc} \bigwedge^c V & \longrightarrow & \cdots & \longrightarrow & \bigwedge^2 V & \xrightarrow{\iota_f} & V \xrightarrow{\iota_f} A \\ & & & & \| & & \\ & & & & \bigoplus_{i=1}^c A\varepsilon_i & & \end{array} \right) \xrightarrow{\sim} A/I \tag{25a}$$

$$\Omega^1_{K/A} = \left(\begin{array}{ccccccc} \bigwedge^c V \otimes W & \longrightarrow & \cdots & \longrightarrow & V \otimes W & \xrightarrow{\iota_f} & W \longrightarrow 0 \\ & & & & \| & & \\ & & & & \bigoplus_{i=1}^c A\,\mathrm{d}\varepsilon_i & & \end{array} \right) = \bigoplus_{i=1}^c K\mathrm{d}\varepsilon_i \tag{25b}$$

$$\mathbb{L}_{(A/I)/A} = \bigoplus_{i=1}^c (A/I)\,\mathrm{d}\varepsilon_i \quad (\text{degree } 1) \;=\; I/I^2[1]. \tag{25c}$$

In (25c) we used $(A/I)^c \simeq I/I^2$; see §1.1.2. Sheafifying, we get $\mathbb{L}_{Z/X} \simeq N^\vee_{Z/X}[1]$.

Note 1.6. *Again, the answer to §1.1.2 is* $\mathrm{Sym}(\mathbb{L}_{Z/X})$. *See Eq.* (7).

Cotangent complex of Fat point. Now we aim to compute $\mathbb{L}_{A/k}$ where $A = k[x]/(x^2)$.

$$K = \left(k[x]\,\varepsilon \xrightarrow{\varepsilon \mapsto x^2} k[x] \right) \xrightarrow{\sim} A \quad \rightsquigarrow \quad \Omega^1_{K/k} = K\mathrm{d}x \oplus K\mathrm{d}\varepsilon \tag{26}$$

with $d(\mathrm{d}\varepsilon) = \mathrm{d}(d\varepsilon) = \mathrm{d}(x^2) = 2x\,\mathrm{d}x$. So

$$\mathbb{L}_{A/k} = \Omega^1_{K/k} \otimes_K A = \left(A\,e \xrightarrow{e \mapsto 2x\mathrm{d}x} A\,\mathrm{d}x \right) \tag{27}$$

Assume $\mathrm{char}\, k \neq 2$. $H_0(\mathbb{L}_{A/k}) = k\,\mathrm{d}x$ has dimension 1, and $H_1(\mathbb{L}_{A/k}) = k\,xe$ has dimension 1. The Euler characteristic $\chi(\mathbb{L}_{A/k}) = 1 - 1 = 0$ matches $\dim A$.

First-order deformations of A over $k[t]/(t^2)$, i.e. square-zero extensions $0 \to tA \to A_t \to A \to 0$ of k-algebras, are classified by $T^1 := \mathrm{Ext}^1_A(\mathbb{L}_{A/k}, A) = \mathrm{Exalcomm}_k(A, A)$. Since both terms of (27) are free, $T^1 = H^1 \mathrm{Hom}_A(\mathbb{L}_{A/k}, A)$, and $\varphi \mapsto \varphi \circ d$ becomes $a \mapsto 2xa$ with $a = \varphi(\mathrm{d}x)$:

$$T^1 = \mathrm{coker}\left(A \xrightarrow{\cdot 2x} A\right) = A/(x) = k. \tag{28}$$

To identify the deformations directly, let A_t be flat over $k[t]/(t^2)$ with $A_t/tA_t = A$, and choose a lift $\tilde{x} \in A_t$ of x. Since $x^2 = 0$ in A, we have $\tilde{x}^2 \in tA$, so $\tilde{x}^2 = t\,g$ for a unique $g \in A$, the amount by which the relation x^2 fails in A_t. The surjection $k[x,t] \to A_t$, $x \mapsto \tilde{x}$, factors through $k[x,t]/(x^2 - t\,g,\, t^2)$, which is free over $k[t]/(t^2)$ on $\{1, x\}$; comparing ranks,

$$A_t \simeq k[x,t]/\big(x^2 - t\,g,\; t^2\big). \tag{29}$$

Changing the lift to $\tilde{x} + t\,a$ replaces g by $g + 2xa$. Hence first-order deformations are classified by $\mathrm{coker}(A \xrightarrow{\cdot 2x} A) = k$, which is (28): in $\mathrm{Hom}_A(\mathbb{L}_{A/k}, A)$, the value $f(e) = g$ on the generator $e = \mathrm{d}\varepsilon$ coming from the relation x^2 records how the relation is deformed, and $h(\mathrm{d}x) = a$ on the generator coming from the coordinate x records the change of lift, with $h \mapsto h \circ d$ being $a \mapsto 2xa$. The generator $g = 1$ gives $A_t = k[x,t]/(x^2 - t,\, t^2)$, nontrivial since no $\tilde{x} = x + t\,a$ satisfies $\tilde{x}^2 = 0$; in the family $x^2 = t$ the double point splits into $x = \pm\sqrt{t}$.

Attempt on non-lci. Consider a non-lci $A = k[x,y]/(x^2, xy)$. Set $R = k[x,y]$. Naively we may try Koszul complex of A:

$$K = \left(R \xrightarrow{(-xy,\, x^2)} R^2 \xrightarrow{(x^2,\, xy)} R \right) \tag{30}$$

$H_0(K) = A$, obviously. However, $H_1(K) \neq 0$, failing to be a resolution. Indeed, using $r = (-y, x)$, we have $\ker(x^2, xy) = Rr$ and $\mathrm{im}(-xy, x^2) = Rxr$, hence $H_1(K) = Rr/(xr) \simeq R/(x) = k[y]$. We conclude that if $f = (f_1, \ldots, f_c)$ is not a regular sequence, we need other further tools to compute $\mathbb{L}_{Z/X}$.

1.2 Loop spaces and Hochschild homology

1.2.1 Derived loop space and Hochschild homology

Goal

Compute the functions on the derived loop space $LX = X \times^h_{X\times X} X$ of $X = \mathbb{A}^n$, i.e. the Hochschild homology $\mathrm{HH}_\bullet(A) = A \otimes^{\mathbb{L}}_{A\otimes A} A$, in two ways: by a Koszul resolution, giving $\mathcal{O}(LX) = \Omega^{-\bullet}_X$ and $LX = T[-1]X$; and by the bar complex, whose comparison with the Koszul resolution yields the coordinate-free HKR isomorphism $\mathrm{HH}_p(A) \simeq \Omega^p_A$.

Function algebra of derived loop space. Given a space X, the derived loop space over it is defined by $LX := X \times^h_{X\times X} X$. For $X = \operatorname{Spec} A$ this means $LX = \operatorname{Spec}(A \otimes^{\mathbb{L}}_{A\otimes A} A)$. We calculate the case $X = \mathbb{A}^n$ to illustrate the relationship between the loop space LX and the Hochschild homology $\mathrm{HH}_\bullet(A) = A \otimes^{\mathbb{L}}_{A\otimes A} A$.

We shall denote $A = k[u_i]$ and $A \otimes_k A = k[x_i, y_i]$. The diagonal map $X \to X \times X$ is equivalent to the $k[x_i, y_i]$-algebra structure on $k[u_i]$ given by $x_i \mapsto u_i, y_i \mapsto u_i$. Then, denoting $V = \bigoplus_{a=1}^n k[x_i, y_i] e_a$, the relevant Koszul resolution of $k[u_i]$ is

$$\bigwedge^n V \longrightarrow \cdots \longrightarrow \bigwedge^3 V \xrightarrow{\iota_f} \bigwedge^2 V \xrightarrow{\iota_f} V \xrightarrow{\iota_f} k[x_i, y_i] \tag{31}$$

where ι_f is defined as in Eq. (3) with $f_i = x_i - y_i$. Tensoring with $k[u_i]$ one obtains a chain complex with vanishing differentials. Hence

$$\mathcal{O}(L\mathbb{A}^n) = k[u_1, \ldots, u_n] \otimes \Lambda(\varepsilon_1, \ldots, \varepsilon_n), \qquad |\varepsilon_i| = 1. \tag{32}$$

Upon identifying $\varepsilon_i \leftrightarrow \mathrm{d}u_i$, this is nothing but the exterior algebra; $\mathcal{O}(L\mathbb{A}^n) \simeq \Omega^{-\bullet}_{\mathbb{A}^n}$.

Shifted tangent bundle. While the result is analogous to that of §1.1.1, here the underlying geometric object is the tangent bundle of X, not a single tangent space, shifted to degree -1. This follows from the observation

$$\Omega^{-\bullet}_X = \mathrm{Sym}_{\mathcal{O}_X}(\Omega^1_X[1]) = \mathrm{Sym}_{\mathcal{O}_X}(T_X[-1]^\vee) = \mathcal{O}(T[-1]X). \tag{33}$$

Hochschild complex. In the following paragraphs, we discuss how the HKR theorem formalizes the identification $\varepsilon_i \leftrightarrow \mathrm{d}u_i$ in a coordinate-free way, via the *Hochschild complex* $C_p(A) = A^{\otimes(p+1)}$ with

$$b(a_0 \otimes \cdots \otimes a_p) = \sum_{i=0}^{p-1} (-1)^i a_0 \otimes \cdots \otimes a_i a_{i+1} \otimes \cdots \otimes a_p + (-1)^p\, a_p a_0 \otimes a_1 \otimes \cdots \otimes a_{p-1}. \quad (34)$$

By definition, the *Hochschild homology* is $\mathrm{HH}_p(A) := H_p(C_\bullet(A), b)$. The homomorphism

$$\Pi : C_p(A) \to \Omega^p_A, \qquad a_0 \otimes \cdots \otimes a_p \mapsto a_0\, \mathrm{d}a_1 \wedge \cdots \wedge \mathrm{d}a_p \quad (35)$$

will induce an isomorphism $\mathrm{HH}_p(A) \xrightarrow{\sim} \Omega^p_A$.

Example 1.7. *One can directly observe* $\mathrm{HH}_1(A) \simeq \Omega^1_A$. *The tail of the Hochshild complex reads*

$$\cdots \longrightarrow A \otimes A \otimes A \xrightarrow{b_2} A \otimes A \xrightarrow{b_1} A \quad (36)$$

First, $b_1(a_0 \otimes a_1) = a_0 a_1 - a_1 a_0 = 0$, *so* $\ker b_1 = A \otimes A$. *On the other hand,*

$$b_2(a_0 \otimes a_1 \otimes a_2) = a_0 a_1 \otimes a_2 - a_0 \otimes a_1 a_2 + a_2 a_0 \otimes a_1. \quad (37)$$

Identifying $a_0 \otimes a_1$ *with* $a_0 \mathrm{d}a_1$, *the quotient by* $\operatorname{im} b_2$ *implements the Leibniz rule.*

Hochschild homology from bar complex. Let $A = k[u_1, \ldots, u_n]$ with $\operatorname{char} k = 0$, and write $A^e := A \otimes A = k[x_i, y_i]$ as above. Our first aim is to see $\mathrm{HH}_\bullet(A) = \mathrm{Tor}^{A^e}_\bullet(A, A)$ using a resolution different from (31). The *bar resolution* of A as A^e-module is $\mathrm{Bar}_p = A \otimes A^{\otimes p} \otimes A$ with $b' = \sum_{i=0}^{p} (-1)^i d_i$, where d_i contracts the i-th and $(i+1)$-th factors:

$$d_i(a_0 \otimes \cdots \otimes a_{p+1}) = a_0 \otimes \cdots \otimes a_i a_{i+1} \otimes \cdots \otimes a_{p+1} \qquad (0 \le i \le p). \quad (38)$$

Exactness follows from the chain homotopy $s(a_0 \otimes \cdots \otimes a_{p+1}) = 1 \otimes a_0 \otimes \cdots \otimes a_{p+1}$.

Proposition 1.8. $b's + sb' = \mathrm{id}$.

Proof. Let $x = a_0 \otimes a_1 \otimes \cdots \otimes a_{p+1}$. Then

$$d_0 s(x) = x \quad (39a)$$

$$d_i s(x) = 1 \otimes a_0 \otimes \cdots \otimes a_{i-1} a_i \otimes \cdots \otimes a_{p+1} \qquad (1 \le i \le p+1) \quad (39b)$$

$$s d_i(x) = 1 \otimes a_0 \otimes \cdots \otimes a_i a_{i+1} \otimes \cdots \otimes a_{p+1} \qquad (0 \le i \le p) \quad (39c)$$

so sd_i and $d_{i+1}s$ cancel in $b's + sb'$ for $0 \le i \le p$. □

Moreover, $A \otimes_{A^e} \mathrm{Bar}_\bullet \simeq C_\bullet(A)$ via $a \otimes (a_0 \otimes v \otimes a_{p+1}) \mapsto a_{p+1} a a_0 \otimes v$, and under this identification the last face d_p of b' becomes the cyclic term of Eq. (34): denoting $v' = a_0 \otimes \cdots \otimes a_{p-1}$ when $v = a_0 \otimes \cdots \otimes a_p$,

$$\begin{array}{ccc} a \otimes (a_0 \otimes v \otimes a_{p+1}) & \longmapsto & a_{p+1} a a_0 \otimes v \\ \big\downarrow{\scriptstyle d_p} & & \vdots \\ a \otimes (a_0 \otimes v' \otimes a_p a_{p+1}) & \longmapsto & a_p a_{p+1} a a_0 \otimes v' \end{array} \quad (40)$$

Hence $\mathrm{HH}_p(A) = \mathrm{Tor}^{A^e}_p(A, A)$.

Comparison with Koszul resolution. Let K be the Koszul resolution (31), with differential ι_f ($f_i = x_i - y_i$). Define an A^e-linear map $\iota : K \to \mathrm{Bar}$ by $\iota_0 = \mathrm{id}_{A^e}$ and

$$\iota(e_{i_1} \wedge \cdots \wedge e_{i_p}) = \sum_{\sigma \in S_p} \mathrm{sgn}(\sigma)\, 1 \otimes u_{i_{\sigma(1)}} \otimes \cdots \otimes u_{i_{\sigma(p)}} \otimes 1. \tag{41}$$

It is a chain map. In $b'\iota(e_{i_1} \wedge \cdots \wedge e_{i_p})$ the inner faces $d_1, \ldots, d_{p-1}$ cancel in pairs $\sigma \leftrightarrow \sigma \circ (j\ j{+}1)$ because A is commutative, while d_0 and d_p produce $x_{i_{\sigma(1)}}$ and $(-1)^p y_{i_{\sigma(p)}}$ times the antisymmetrization of the remaining indices. Collecting terms by the removed index i_j, and using $\mathrm{sgn}(\sigma) = (-1)^{j-1}\mathrm{sgn}(\tau)$ (resp. $(-1)^{p-j}\mathrm{sgn}(\tau)$) when i_j is moved to the front (resp. the end), gives

$$b'\iota(e_{i_1} \wedge \cdots \wedge e_{i_p}) = \sum_{j=1}^{p} (-1)^{j-1}(x_{i_j} - y_{i_j})\, \iota(e_{i_1} \wedge \cdots \widehat{e_{i_j}} \cdots \wedge e_{i_p}) = \iota\big(\iota_f(e_{i_1} \wedge \cdots \wedge e_{i_p})\big).$$

Since ι lifts id_A between two free resolutions, it is a homotopy equivalence. Therefore

$$\iota \otimes_{A^e} A : \ a\, \varepsilon_{i_1} \wedge \cdots \wedge \varepsilon_{i_p} \ \mapsto \ \sum_{\sigma \in S_p} \mathrm{sgn}(\sigma)\, a \otimes u_{i_{\sigma(1)}} \otimes \cdots \otimes u_{i_{\sigma(p)}} \tag{42}$$

is a quasi-isomorphism from the complex Eq. (32), whose differential vanishes, to $(C_\bullet(A), b)$.

Hochschild–Kostant–Rosenberg. For $x = a_0 \otimes \cdots \otimes a_{p+1}$ put $T_j = a_0 a_j\, \mathrm{d}a_1 \wedge \cdots \widehat{\mathrm{d}a_j} \cdots \wedge \mathrm{d}a_{p+1}$. The Leibniz rule gives $\Pi(d_0 x) = T_1$, $\Pi(d_i x) = T_i + T_{i+1}$ for $1 \le i \le p$, and $\Pi(d_{p+1}x) = T_{p+1}$, so $\Pi(bx) = \sum_i (-1)^i \Pi(d_i x) = 0$ by telescoping. Thus Π induces a map $\mathrm{HH}_p(A) \to \Omega^p_A$.

Composing the maps, $\Pi \circ (\iota \otimes_{A^e} A)(a\, \varepsilon_{i_1} \wedge \cdots \wedge \varepsilon_{i_p}) = p!\, a\, \mathrm{d}u_{i_1} \wedge \cdots \wedge \mathrm{d}u_{i_p}$. Since $\varepsilon_I \mapsto \mathrm{d}u_I$ is an isomorphism $\bigwedge^p A^n \simeq \Omega^p_A$ and Eq. (42) is a quasi-isomorphism, Π induces an isomorphism

$$\mathrm{HH}_p(A) \ \xrightarrow{\sim} \ \Omega^p_A. \tag{43}$$

Since Π is defined without coordinates, the isomorphism Eq. (43) does not depend on the choice of the u_i.

1.2.2 Hochschild homology of projective line

Goal

Extend Hochschild homology to the non-affine $\mathbb{P}^1$ by gluing the Hochschild complexes of a two-chart cover, and prove $\mathrm{HH}_n(\mathbb{P}^1) = \check{H}^{-n}(\mathcal{O}) \oplus \check{H}^{1-n}(\Omega^1)$ from the Mayer–Vietoris sequence and chart-wise HKR. Answer: $\mathrm{HH}_0(\mathbb{P}^1) = k \cdot 1 \oplus k \cdot [t^{-1}\mathrm{d}t]$ and $\mathrm{HH}_{n\neq 0}(\mathbb{P}^1) = 0$; the second class exists only because of the gluing.

For a non-affine space X, the Hochschild complex of algebras should be upgraded to a complex of sheaves of algebras. Its hypercohomology captures the differential forms $\Omega^\bullet_X$.

$$\mathrm{HH}_n(\mathbb{P}^1) = \check{H}^{-n}(\mathcal{O}_{\mathbb{P}^1}) \oplus \check{H}^{1-n}(\Omega^1_{\mathbb{P}^1}) \tag{44}$$

Here $\mathrm{HH}_\bullet(\mathbb{P}^1)$ is defined by gluing the Hochschild complexes of an affine cover. We use the cover $\mathcal{U} = \{U_0, U_1\}$ with $U_0 = \operatorname{Spec} k[t]$, $U_1 = \operatorname{Spec} k[s]$, $s = t^{-1}$. Put $C^{p,q} = \check{C}^p(\mathcal{U}, C_q(\mathcal{O}))$, i.e.,

$$C^{0,q} = C_q(k[t]) \oplus C_q(k[s]), \qquad C^{1,q} = C_q(k[t, t^{-1}]), \tag{45}$$

with the Hochschild differential b of Eq. (34) in the q-direction and the Čech differential $\delta(c_0, c_1) = c_1|_{U_{01}} - c_0|_{U_{01}}$ in the p-direction. Since the restrictions are ring homomorphisms, b commutes with δ, and $D = b + (-1)^q\delta$ squares to zero on the total complex $\mathrm{Tot}_n = \bigoplus_{q-p=n} C^{p,q}$. The Čech degree is cohomological, so it lowers the total homological degree. We define

$$\mathrm{HH}_n(\mathbb{P}^1) := H_n\big(\mathrm{Tot}\ C^{\bullet,\bullet}, D\big). \tag{46}$$

This is the hypercohomology of the sheafified Hochschild complex, and it does not depend on the affine cover [9]. Taking global sections first would give $\mathrm{HH}_\bullet(\mathcal{O}(\mathbb{P}^1)) = \mathrm{HH}_\bullet(k) = k$, which misses the class $[t^{-1}\mathrm{d}t]$ found below.

Ungluing by Mayer–Vietoris. An element of Tot_n is a triple

$$(c_0, c_1; c_{01}) \in C_n(k[t]) \oplus C_n(k[s]) \oplus C_{n+1}(k[t, t^{-1}]), \tag{47}$$

and the total differential reads

$$D(c_0, c_1; c_{01}) = \big(bc_0,\ bc_1;\ bc_{01} + (-1)^n(c_1 - c_0)|_{U_{01}}\big). \tag{48}$$

The low-degree part of the double complex looks like

$$\begin{array}{lcc}
 & \vdots & \vdots \\
 & \downarrow b & \downarrow b \\
q = 2 & C_2(k[t]) \oplus C_2(k[s]) & \xrightarrow{(-1)^q\delta} C_2(k[t, t^{-1}]) \\
 & \downarrow b \quad \searrow \mathrm{Tot}_1 & \downarrow b \\
q = 1 & C_1(k[t]) \oplus C_1(k[s]) & \xrightarrow{(-1)^q\delta} C_1(k[t, t^{-1}]) \\
 & \downarrow b=0 \quad \searrow \mathrm{Tot}_0 & \downarrow b=0 \\
q = 0 & k[t] \oplus k[s] & \xrightarrow{(-1)^q\delta} k[t, t^{-1}] \\
 & p = 0 & p = 1
\end{array} \tag{49}$$

Using $C_0(R) = R$ and $b = 0$ on $C_1(R)$ for a commutative ring R, the total complex ends with

$$\cdots \to \mathrm{Tot}_1 \xrightarrow{D_1} \underbrace{k[t] \oplus k[s] \oplus C_1(k[t, t^{-1}])}_{\mathrm{Tot}_0} \xrightarrow{D_0} \underbrace{k[t, t^{-1}]}_{\mathrm{Tot}_{-1}} \to 0, \quad D_0(f_0, f_1; c_{01}) = f_1 - f_0. \tag{50}$$

For instance, $(1,1;0)$ and $(0,0;t^{-1}\otimes t)$ are cycles in Tot_0, since $1-1=0$ and $b(t^{-1}\otimes t)=t^{-1}t-t\,t^{-1}=0$. They will represent 1 and $[t^{-1}\mathrm{d}t]$ in Eq. (59).

The triples $(0,0;c_{01})$ form a subcomplex, namely the column $p=1$, which is a copy of $C_{\bullet+1}(k[t,t^{-1}])$; the quotient is the column $p=0$, namely $(C_\bullet(k[t])\oplus C_\bullet(k[s]),b)$. Write $\mathrm{HH}_n(U)$ for $\mathrm{HH}_n(\mathcal{O}(U))$ and $\delta_n:\mathrm{HH}_n(U_0)\oplus\mathrm{HH}_n(U_1)\to\mathrm{HH}_n(U_{01})$, $(\alpha_0,\alpha_1)\mapsto\pm(\alpha_1-\alpha_0)|_{U_{01}}$. The long exact sequence of this short exact sequence of complexes is the Mayer–Vietoris sequence

$$\begin{aligned}\cdots\to\mathrm{HH}_{n+1}(U_0)\oplus\mathrm{HH}_{n+1}(U_1)\xrightarrow{\delta_{n+1}}\mathrm{HH}_{n+1}(U_{01})\to\mathrm{HH}_n(\mathbb{P}^1)\\ \to\mathrm{HH}_n(U_0)\oplus\mathrm{HH}_n(U_1)\xrightarrow{\delta_n}\mathrm{HH}_n(U_{01})\to\mathrm{HH}_{n-1}(\mathbb{P}^1)\to\cdots.\end{aligned}\tag{51}$$

Its connecting map is δ_n, because for Hochschild cycles c_0,c_1, Eq. (48) gives $D(c_0,c_1;0)=(0,0;\pm(c_1-c_0))$. Equivalently, for each n,

$$0\to\operatorname{coker}\delta_{n+1}\to\mathrm{HH}_n(\mathbb{P}^1)\to\ker\delta_n\to0.\tag{52}$$

HKR on each chart. Each of the three rings $R=k[t]$, $k[s]$, $k[t,t^{-1}]$ has a global coordinate $w=t$, s, t, and

$$\mathrm{HH}_0(R)=R,\qquad \mathrm{HH}_1(R)=\{[a\otimes w]:a\in R\}\xrightarrow[\sim]{\Pi}\Omega^1_R=R\,\mathrm{d}w,\qquad \mathrm{HH}_{\geq2}(R)=0,\tag{53}$$

with $\Pi[a\otimes w]=a\,\mathrm{d}w$. For $k[t]$ and $k[s]$ this is Eq. (43). For $k[t,t^{-1}]$ the proof of Eq. (43) goes through verbatim. With $A^e=k[x^{\pm1},y^{\pm1}]$, the sequence $0\to A^e e\xrightarrow{x-y}A^e\to k[t,t^{-1}]\to0$ is exact, because $x-y$ is a nonzerodivisor in the domain A^e and $A^e/(x-y)\simeq k[t,t^{-1}]$; the map $\iota(e)=1\otimes t\otimes1$ is a chain map, because $b'(1\otimes t\otimes1)=t\otimes1-1\otimes t=x-y$; and $\Omega^1_{k[t,t^{-1}]}=k[t,t^{-1}]\,\mathrm{d}t$ is free. For instance $[t^{-1}\otimes t]\in\mathrm{HH}_1(k[t,t^{-1}])$ maps to $t^{-1}\mathrm{d}t$.

Reduction to differential forms. The identity $\Pi\circ b=0$ says that Π is a chain map from $(C_\bullet(R),b)$ to $(\Omega^\bullet_R,0)$, the forms with zero differential, and by HKR it is a quasi-isomorphism on each chart, so $\mathrm{HH}_n(R)\simeq\Omega^n_R$. Since Π commutes with restrictions, δ_n becomes the Čech map $d_0:\Omega^n(U_0)\oplus\Omega^n(U_1)\to\Omega^n(U_{01})$. For a cover by two sets this is the whole Čech complex, so $\ker\delta_n=\check H^0(\Omega^n)$, $\operatorname{coker}\delta_{n+1}=\check H^1(\Omega^{n+1})$, and $\check H^p=0$ for $p\neq0,1$.

The sequence Eq. (52) splits canonically. Applying Π to all three components maps Tot to Tot_Ω, the same construction with $(\Omega^\bullet,0)$ in place of $(C_\bullet,b)$. As the internal differential vanishes, the only differential of Tot_Ω is the Čech one, which preserves the form degree q; hence Tot_Ω is the direct sum

$$\mathrm{Tot}^n_\Omega=\bigoplus_{q-p=n}\check C^p(\Omega^q)\tag{54}$$

Concretely, a Čech 0-cocycle (ω_0,ω_1) of Ω^n is already a cycle $(\omega_0,\omega_1;0)$ in Tot^n_Ω, whereas in Tot a n-cycle is given by the datum $(c_0,c_1;c_{01})$ of a pair of Hochschild cycles and a choice of c_{01} with $bc_{01}=\pm(c_1-c_0)$, unique only up to a Hochschild cycle, that is, up to the image of $\operatorname{coker}\delta_{n+1}$.

Gluing back. The Mayer–Vietoris sequences of Tot and Tot_Ω agree term by term via the chart-wise HKR isomorphisms, so the five lemma shows that Π is a quasi-isomorphism, and

$$\mathrm{HH}_n(\mathbb{P}^1) \simeq H_n(\mathrm{Tot}_\Omega) = \check{H}^0(\Omega^n) \oplus \check{H}^1(\Omega^{n+1}) = \bigoplus_{\substack{p,q\in\{0,1\}\\ q-p=n}} \check{H}^p(\Omega^q) = \check{H}^{-n}(\mathcal{O}) \oplus \check{H}^{1-n}(\Omega^1). \quad (55)$$

This is Eq. (44). For a smooth X over k of characteristic 0, the same argument with local étale coordinates and more charts gives $\mathrm{HH}_n(X) \simeq \bigoplus_q H^{q-n}(X, \Omega_X^q)$. It remains to compute the Čech maps d_0 for $\mathcal{O}$ and for Ω^1, which we do next.

Čech complex of $\mathcal{O}$. $\mathbb{P}^1$ is covered by two lines $U_0 = \operatorname{Spec} k[t]$ and $U_1 = \operatorname{Spec} k[s]$, related by $s = t^{-1}$, with overlap $U_0 \cap U_1 = \operatorname{Spec} k[t, t^{-1}]$. The corresponding Čech complex is

$$k[t] \oplus k[s] \xrightarrow{d_0} k[t, t^{-1}], \qquad d_0(f_0, f_1) = f_1 - f_0. \quad (56)$$

It is straightforward that $\check{H}^0(\mathcal{O}) = \ker d_0 \simeq k$ and $\check{H}^1(\mathcal{O}) = \operatorname{coker} d_0 = 0$.

Čech complex of Ω^1. The Čech complex is

$$k[t]\,\mathrm{d}t \oplus k[s]\,\mathrm{d}s \xrightarrow{d_0} k[t, t^{-1}]\,\mathrm{d}t \quad (57)$$

with $\mathrm{d}s = -t^{-2}\mathrm{d}t$. The map d_0 is explicitly

$$d_0(f_0(t)\,\mathrm{d}t, f_1(s)\,\mathrm{d}s) = \big(-t^{-2} f_1(t^{-1}) - f_0(t)\big)\,\mathrm{d}t, \quad (58)$$

hence $\check{H}^0(\Omega^1) = \ker d_0 = 0$ and $\check{H}^1(\Omega^1) = \operatorname{coker} d_1 = k \cdot [t^{-1}\mathrm{d}t]$.

Combining the results,

$$\mathrm{HH}_0(\mathbb{P}^1) = k \cdot 1 \oplus k \cdot [t^{-1}\mathrm{d}t], \qquad \mathrm{HH}_{n\neq 0}(\mathbb{P}^1) = 0. \quad (59)$$

Negative degrees are allowed in Eq. (44); here $\mathrm{HH}_{-1}(\mathbb{P}^1) = \check{H}^1(\mathcal{O}) = 0$.

1.2.3 Loops in BG

Goal (planned)

For a finite group G, compute $\mathrm{pt} \times^h_{BG} \mathrm{pt} \simeq G$ and $LBG \simeq [G/G] = \bigsqcup_{[g]} BC(g)$, and match it with the Hochschild homology of the group algebra: $\mathrm{HH}_\bullet(k[G]) \simeq \bigoplus_{[g]} H_\bullet(C(g); k) = H_\bullet(LBG; k)$ (Burghelea), the group-algebra counterpart of HKR.

Planned content. [Guide §5, Steps 1–2; HH note §9] For a finite group G, the based loops are $\mathrm{pt} \times^h_{BG} \mathrm{pt} \simeq G$, and $LBG = \mathrm{Map}(S^1, BG) \simeq [G/G] = \bigsqcup_{[g]} BC(g)$, whose functions are the class functions. On the algebra side, the holonomy decomposition of the Hochschild complex gives $\mathrm{HH}_0(k[G]) = k^{\{\text{conjugacy classes}\}}$ and, by Burghelea, $\mathrm{HH}_\bullet(k[G]) \simeq \bigoplus_{[g]} H_\bullet(C(g); k) = H_\bullet(LBG; k)$, the group-algebra counterpart of Eq. (32).

1.3 Derived categories and noncommutative geometry

1.3.1 Hochschild homology via Beilinson

Goal

Recompute $\mathrm{HH}_\bullet(\mathbb{P}^1)$ without gluing. Beilinson's tilting bundle $T = \mathcal{O} \oplus \mathcal{O}(1)$ identifies $D^b(\mathsf{Coh}\,\mathbb{P}^1)$ with modules over the Kronecker quiver algebra $A = \mathrm{End}(T)$, and a two-term bimodule resolution gives $\mathrm{HH}_0(A) = k^2$, $\mathrm{HH}_{\geq 1}(A) = 0$. The two answers match class by class: the glued class $[t^{-1}\mathrm{d}t]$ corresponds to the second vertex of the quiver.

In §1.2.2, non-affine nature of $\mathbb{P}^1$ necessitates hypercohomology. Non-commutative geometry offers a different route: Beilinson's theorem [2] says $\mathbb{P}^1$ is "affine", in the sense that

$$\mathbb{R}\mathrm{Hom}(T, -) : D^b(\mathsf{Coh}\,\mathbb{P}^1) \xrightarrow{\sim} D^b(\mathsf{mod}\,A), \qquad T = \mathcal{O} \oplus \mathcal{O}(1), \quad A = \mathrm{End}(T). \tag{60}$$

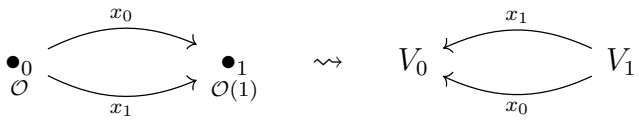
To find A we use $\mathrm{Hom}(\mathcal{O}(i), \mathcal{O}(j)) = H^0(\mathcal{O}(j-i))$ and $\mathrm{Ext}^1(\mathcal{O}(i), \mathcal{O}(j)) = H^1(\mathcal{O}(j-i))$.

	$\mathrm{Hom} = H^0(\mathcal{O}(j-i))$	$\mathrm{Ext}^1 = H^1(\mathcal{O}(j-i))$
$\mathcal{O} \to \mathcal{O}$, $\mathcal{O}(1) \to \mathcal{O}(1)$	k	$H^1(\mathcal{O}) = 0$
$\mathcal{O} \to \mathcal{O}(1)$	$H^0(\mathcal{O}(1)) = k\,x_0 \oplus k\,x_1$	$H^1(\mathcal{O}(1)) = 0$
$\mathcal{O}(1) \to \mathcal{O}$	$H^0(\mathcal{O}(-1)) = 0$	$H^1(\mathcal{O}(-1)) = 0$

$A = k\,e_0 \oplus k\,e_1 \oplus k\,x_0 \oplus k\,x_1$ where e_i is the projector onto $\mathcal{O}(i)$ and the maps $x_i : T \to T$ are restricted to $\chi_i : \mathcal{O} \to \mathcal{O}(1)$. The relations $e_i^2 = e_i$, $e_0 + e_1 = 1$, $e_0 e_1 = e_1 e_0 = 0$, $x_i = e_1 x_i = x_i e_0 = e_1 x_i e_0$, $x_i x_j = e_0 x_i = x_i e_1 = 0$ can be read from the matrix representation adapted to the splitting $T = \mathcal{O} \oplus \mathcal{O}(1)$:

$$e_0 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad e_1 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \quad x_0 = \begin{bmatrix} 0 & 0 \\ \chi_0 & 0 \end{bmatrix}, \quad x_1 = \begin{bmatrix} 0 & 0 \\ \chi_1 & 0 \end{bmatrix}. \tag{61}$$

In the quiver representation:



(62)

The right side of Eq. (62) represents a right A-module $M = \mathrm{Hom}(T, F)$, where $F \in \mathsf{Coh}\,\mathbb{P}^1$. The vertices are $V_i = Me_i$. Concretely, $V_0 = \mathrm{Hom}(\mathcal{O}, F) = H^0(F)$ and $V_1 = \mathrm{Hom}(\mathcal{O}(1), F) = H^0(F(-1))$, and x_i acts by precomposition with χ_i, i.e. by the multiplication $\chi_i : H^0(F(-1)) \to H^0(F)$. For example, $\mathcal{O} \mapsto e_0 A$ with $(V_0, V_1) = (k, 0)$, and $\mathcal{O}(1) \mapsto e_1 A$ with $(V_0, V_1) = (H^0(\mathcal{O}(1)), H^0(\mathcal{O})) = (k^2, k)$ and $x_i : 1 \mapsto x_i$.

Using Beilinson's theorem, the Hochschild homology of $\mathbb{P}^1$ is just $\mathrm{HH}_\bullet(A)$.

Remark 1.9. *This relies on two facts: the Hochschild homology of a scheme agrees with that of its dg category of perfect complexes [7], and Hochschild homology is invariant under derived equivalence [6]. Since* $\mathbb{P}^1$ *is smooth,* $D^b(\mathsf{Coh}\,\mathbb{P}^1) = \mathsf{Perf}(\mathbb{P}^1)$*, and Eq. (60) restricts to* $\mathsf{Perf}(\mathbb{P}^1) \simeq \mathsf{Perf}(A)$.

It is straightforward that $\mathrm{HH}_0(A) = A/[A,A] = k\,e_0 \oplus k\,e_1$, since $[e_1, x_i] = x_i$ while e_0, e_1 are not commutators. In higher degrees, use the (A,A)-bimodule resolution of a path algebra.

$$0 \longrightarrow \bigoplus_{a:i\to j} Ae_j \otimes e_i A \longrightarrow \bigoplus_{v} Ae_v \otimes e_v A \longrightarrow A \longrightarrow 0 \tag{63a}$$

$$\overset{A\otimes_{A^e} -}{\rightsquigarrow} \quad 0 \longrightarrow \bigoplus_{a:i\to j} e_i A e_j \longrightarrow \bigoplus_{v} e_v A e_v \longrightarrow 0 \tag{63b}$$

In Eq. (63a), the generator $e_j \otimes e_i$ of the summand of an arrow $a : i \to j$ (so $a = e_j a e_i$) maps to $a \otimes e_i - e_j \otimes a$, and the last map is multiplication; exactness is standard [3]. Since the quiver (62) has no cycles, $e_v A e_v = k\,e_v$ and $e_i A e_j = 0$. Thus, Eq. (59) is reproduced:

$$\mathrm{HH}_0(A) \simeq k^2, \qquad \mathrm{HH}_{\geq 1}(A) = 0. \tag{64}$$

Matching the two computations. The two answers agree as classes, not only in dimension. Under Eq. (60), $\mathcal{O} \mapsto e_0 A$ and $\mathcal{O}(1) \mapsto e_1 A$, and the Chern character of a projective module eA in $\mathrm{HH}_0(A) = A/[A,A]$ is the class $[e]$. On the Čech side, $\mathcal{O}(1)$ is trivialized by x_0 on U_0 and by $x_1 = t\,x_0$ on U_1, so its transition function on U_{01} is t, and $c_1(\mathcal{O}(1))$ is the Čech class of $\mathrm{d}\log t = t^{-1}\mathrm{d}t$. Equivalently, the Dennis trace of the unit t is the Hochschild cycle $t^{-1} \otimes t$ on U_{01}, with $\Pi(t^{-1} \otimes t) = t^{-1}\mathrm{d}t$. Hence $\mathrm{ch}(\mathcal{O}) = 1$ and $\mathrm{ch}(\mathcal{O}(1)) = 1 + [t^{-1}\mathrm{d}t]$ (up to a sign convention), and the bases of Eq. (59) and Eq. (64) correspond as

$$1 \leftrightarrow [e_0], \qquad [t^{-1}\mathrm{d}t] \leftrightarrow [e_1] - [e_0]. \tag{65}$$

The class that the Čech computation produces by gluing two charts appears on the Beilinson side as the second vertex of the quiver. As a check, the residue $\mathrm{Res}_{t=0}(t^{-1}\mathrm{d}t) = 1 = \deg \mathcal{O}(1)$.

1.3.2 Perfect complexes and Morita invariance

Goal (planned)

Define perfect complexes and characterize them as compact objects; show with the dual numbers that $\mathsf{Perf}(A) \subsetneq D^b(\mathsf{mod}\,A)$ in general, with equality in finite global dimension. State the results used in §1.3.1: Bondal–Van den Bergh and the derived Morita invariance of HH.

Planned content. [Not in the Guide] Promote Remark 1.9 to the main text. Define $\mathsf{Perf}(A)$ as the complexes quasi-isomorphic to bounded complexes of finitely generated projectives, and $\mathsf{Perf}(X)$ by the same condition locally; characterize both as the compact objects of $D(A)$ and $D_{\mathrm{qc}}(X)$. Show $\mathsf{Perf}(A) \subsetneq D^b(\mathsf{mod}\,A)$ in general, using k over $k[x]/(x^2)$ with its infinite periodic resolution, and equality in finite global dimension, e.g. for $\mathbb{P}^1$ and the Kronecker algebra. State Bondal–Van den Bergh (a compact generator G gives $D_{\mathrm{qc}}(X) \simeq D(\mathrm{REnd}\,G)$) and Keller's derived Morita invariance of HH.

1.3.3 Centers and loop spaces

Goal (planned)

Upgrade $\mathrm{HH}(\mathsf{QC}(X)) = \mathcal{O}(LX)$ to $Z(\mathsf{QC}(X)) \simeq \mathsf{QC}(LX)$ and specialize to $X = BG$: the simple objects $([g], \rho)$ of $Z(\mathsf{Rep}\, G) \simeq \mathsf{QC}(G/G)$ are the anyons of the Drinfeld double $D(G)$, and twisting by $\omega \in H^3(G; U(1))$ gives $D^\omega(G)$ (double semion). This is the entry point to the SymTFT of a finite symmetry.

Planned content. [Guide §5, Steps 3–4] $\mathrm{HH}(\mathsf{QC}(X)) \simeq \mathcal{O}(LX)$ and $Z(\mathsf{QC}(X)) \simeq \mathsf{QC}(LX)$ (Ben-Zvi–Francis–Nadler). For $X = BG$ this gives $Z(\mathsf{Rep}\, G) \simeq \mathsf{QC}(G/G)$, whose simple objects $([g], \rho)$ with $\rho \in \operatorname{Irr} C(g)$ are the anyons of the Drinfeld double $D(G)$. Twisting by $\omega \in H^3(G; U(1))$ through transgression gives $D^\omega(G)$; worked example: the $\mathbb{Z}_2$ double semion. This is the entry point to the SymTFT of a finite symmetry, used in §4.1.

1.4 Derived critical loci and BV

Throughout, $X = \mathbb{A}^n = \operatorname{Spec} k[x_1, \dots, x_n]$, $\operatorname{char} k = 0$, $f \in \mathcal{O}(X)$, and degrees are cohomological, so the antifields ξ_i sit in degree -1.

1.4.1 Derived critical locus as a derived intersection

Goal

Compute $\mathrm{dCrit}(f) = X \times^h_{T^*X} X$, the derived intersection of the zero section and the graph of df, by resolving one side with a Koszul complex as in §1.1.1:

$$\mathcal{O}(\mathrm{dCrit}\, f) = \big(k[x] \otimes \Lambda(\xi_1, \dots, \xi_n),\ d\xi_i = \partial_i f\big) = \big(\textstyle\bigwedge^{-\bullet} T_X,\ \iota_{df}\big).$$

Read off $H^0 = k[x]/(\partial f) = \mathcal{O}(\mathrm{Crit}\, f)$ and $H^{-1} = \{v : v(f) = 0\}/\iota_{df}(\bigwedge^2 T_X)$, and check $\mathrm{dCrit}(0) = T^*[-1]X$.

The Koszul algebra that resolves $\mathcal{O}(s_0) = k[x_i, p_i]/(p_i)$ is $K = k[x_i, p_i][\xi_i]$, with differential $d\xi_i = p_i$. Tensoring with $\mathcal{O}(\Gamma_{df}) = k[x_i, p_i]/(p_i - \partial_i f)$, one gets $\mathcal{O}(\mathrm{dCrit}\, f) = k[x_i][\xi_i]$ with $d\xi_i = \partial_i f$.

$$k[x_i] \otimes \Lambda^n(\xi_i) \xrightarrow{d} \cdots \xrightarrow{d} k[x_i] \otimes \Lambda^1(\xi_i) \xrightarrow{d} k[x_i] \tag{66}$$

It is straightforward that $H^0 = k[x_i]/(\partial_i f)$. At cohomological degree -1, $\ker d$ consists of those polynomials $\sum_j v^j(x)\xi_j$ such that $v^j(x)\partial_j f = v(f) = 0$ where $v = \sum_j v^j \partial_j$.

Polyvector fields. The generators ξ_i have a geometric meaning. The ideal of the zero section $s_0(X) \subset T^*X$ is generated by the fiber coordinates p_i, and p_i is the function $(x, \alpha) \mapsto \alpha(\partial_i)$, linear on the fibers. Linear functions on the fibers of T^*X are sections of $(T^*X)^\vee = T_X$, so the conormal module of the zero section is

$$I/I^2 \simeq T_X, \qquad p_i \mapsto \partial_i, \tag{67}$$

and the Koszul resolution is $\bigwedge^\bullet$ of it, as in §1.1.2. Concretely, under a change of coordinates $x \mapsto x'$ we have $p_i = \sum_j (\partial x'_j/\partial x_i)\, p'_j$; since $d\xi_i = p_i$, the ξ_i transform in the same way, which is the transformation law of ∂_i. Hence

$$k[x_i] \otimes \Lambda^p(\xi_1, \dots, \xi_n) \xrightarrow{\sim} \bigwedge^p T_X, \qquad \xi_{i_1} \cdots \xi_{i_p} \mapsto \partial_{i_1} \wedge \cdots \wedge \partial_{i_p}, \tag{68}$$

placed in cohomological degree $-p$, does not depend on the coordinates. Under this identification $d\partial_i = \partial_i f = \langle df, \partial_i \rangle$, and since d is a derivation of odd degree,

$$d(v_1 \wedge \cdots \wedge v_p) = \sum_{k=1}^{p} (-1)^{k-1} \langle df, v_k \rangle\, v_1 \wedge \cdots \widehat{v_k} \cdots \wedge v_p, \tag{69}$$

which is the contraction ι_{df}. Therefore

$$\mathcal{O}(\mathrm{dCrit}\, f) \simeq \big(\textstyle\bigwedge^{-\bullet} T_X,\ \iota_{df} \big), \tag{70}$$

and since both sides are defined without coordinates, (70) holds for any smooth X as an equivalence of sheaves of CDGAs. In particular, In degree -1 $\operatorname{im} d$ consists of

$$\iota_{df}(\partial_i \wedge \partial_j) = \partial_i f\, \partial_j - \partial_j f\, \partial_i, \tag{71}$$

vector fields that annihilate f for the trivial reason of antisymmetry.

Example 1.10. *For $f = 0$, the differential $d = 0$ is trivial and*

$$\mathcal{O}(\mathrm{dCrit}\, 0) = \textstyle\bigwedge^{-\bullet} T_X = \mathrm{Sym}_{\mathcal{O}_X}\big(T_X[1]\big) = \mathcal{O}\big(T^*[-1]X\big), \tag{72}$$

the functions on the shifted cotangent bundle $T^*[-1]X$, whose fiber coordinates ξ_i have cohomological degree -1. Compare the derived loop space $LX = T[-1]X$ of §1.2, whose fiber coordinates $\mathrm{d}x_i$ have degree -1: there the tangent bundle is shifted, here the cotangent bundle.

1.4.2 Isolated critical points

Goal

Show that at an isolated critical point $(\partial_1 f, \dots, \partial_n f)$ is a regular sequence, so dCrit f carries no derived information there: $H^{<0} = 0$ and H^0 is the Milnor algebra, of dimension the Milnor number μ. Compute $\mu = 1$ for Morse and $\mu = k$ for A_k, $f = x^{k+1}/(k+1)$. Show that the Morsification $x^3/3 - \varepsilon x$ of $f = x^3/3$ is the first-order deformation $x^2 - \varepsilon$ generating $T^1 = k$ of the fat point in §1.1.5.

In a smooth space, $\mathcal{O}_p$ is a regular local ring, hence local Cohen–Macaulay. Then $(\partial_1 f, \dots, \partial_n f)$ is a regular sequence if and only if $\mathcal{O}_p/(\partial_1 f, \dots, \partial_n f)$ is zero-dimensional, which is satisfied at an isolated critical point.

Locally, take an isomorphism $p \in U \simeq \operatorname{Spec} R$. By the discussions in §1.1.2, (∂f) being a regular sequence guarantees that the Koszul complex $R[\theta_1, \dots, \theta_n]$ with $d\theta_i = \partial_i f$ has $H^{<0}(K) = 0$. This means that dCrit f is locally isomorphic (as a derived scheme) to $\operatorname{Spec} H^0(K) = \operatorname{Crit} f$. Here $H^0(K)$ is the Milnor algebra with Milnor number $\mu = \dim_k H^0(K)$.

Example 1.11. *Take $k = \mathbb{C}$. At a Morse critical point, $f = x_1^2 + \cdots + x_n^2$ locally. Then*

$$R = \mathbb{C}[x_i]_{(2x_i)}, \qquad H^0(K) = R/(2x_i) \simeq \mathbb{C} \quad \Longrightarrow \quad \mu = 1. \tag{73}$$

Example 1.12. *On a A_k singularity associated to $f = x^{k+1}/(k+1)$ on $\mathbb{A}^1$,*

$$H^0(K) = \mathbb{C}[x]/(x^k) \simeq \mathbb{C}^k \quad \Longrightarrow \quad \mu = k. \tag{74}$$

$f = x^3/3$ has a Morsification $x^3/3 - \varepsilon x$ which gives $H^0(K) = \mathbb{C}[x]/(x^2 - \varepsilon)$, a family connecting a pair of simple points to a double point.

1.4.3 Tangent complex and the (-1)-shifted symplectic structure

Goal (planned)

Compute $\mathbb{T}_{\mathrm{dCrit}\, f} = \left[T_X \xrightarrow{\mathrm{Hess}\, f} T_X^*\right]$ in degrees $0, 1$, both from the fiber product and from the Koszul model as in §1.1.5. Deduce $\mathbb{T} \simeq \mathbb{T}^\vee[-1]$ from the symmetry of the Hessian, identify it with $\omega = \sum_i dx_i \wedge d\xi_i$ of degree -1, and note virtual dimension 0. For $f = x^3/3$, recover the fat point's $(T^0, T^1) = (k\, x\partial_x, k)$.

Planned content. [BV guide §3] $\mathbb{T}_{X \times^h_Z Y} = \mathrm{fib}(\mathbb{T}_X \oplus \mathbb{T}_Y \to \mathbb{T}_Z)$ with $ds_0(v) = (v, 0)$, $d\Gamma_{df}(w) = (w, \mathrm{Hess}(f)w)$. In the Koszul model $d(\mathrm{d}\xi_i) = \sum_j \partial_i \partial_j f \, \mathrm{d}x_j$; for $f = x^3/3$ the model $d\xi = x^2$ is literally the resolution of §1.1.5 with $\xi = \varepsilon$. At $p \in \mathrm{Crit}\, f$: $H^0 = \ker \mathrm{Hess}_p$, $H^1 = \mathrm{coker}\, \mathrm{Hess}_p$. Exercise: $f = x^2 y$ at $(0, y_0 \neq 0)$ and at the origin. Cite PTVV: a Lagrangian intersection in a 0-shifted symplectic space is (-1)-shifted symplectic.

1.4.4 BV dictionary

Goal (planned)

Translate §1.4.1 into BV language: antifields $\xi_i = x_i^+$, antibracket $\{x_i, \xi_j\} = \delta_{ij}$ (the Schouten bracket), $Q = \{f, -\} = \iota_{df}$, and the master equation $\{f, f\} = 0$. Show that the BV Laplacian $\Delta = \sum_i \partial_{x_i} \partial_{\xi_i}$ is the divergence, generates the bracket, and becomes the de Rham differential under polyvectors $\simeq$ forms via $\mathrm{d}x_1 \cdots \mathrm{d}x_n$, so that $\iota_{df} + \hbar\Delta$ is the twisted de Rham differential $df \wedge + \hbar\, \mathrm{d}$.

Planned content. [BV guide §4] Check $\Delta(ab) = (\Delta a)b + (-1)^{|a|} a \Delta b + (-1)^{|a|}\{a, b\}$ on $a = x_i$, $b = \xi_i$. Exercise: $\mathrm{div}(x\partial_x - 2y\partial_y) = -1$, so the $\mathbb{G}_m$-symmetry of $f = x^2 y$ in §1.4.5 does not preserve $\mathrm{d}x\, \mathrm{d}y$ (a measure anomaly). Cite: for isolated tame f, the twisted de Rham cohomology has rank μ, recovering §1.4.2.

1.4.5 Flat directions and gauge symmetries

Goal (planned)

Compute examples with $H^{-1} \neq 0$. For $f = x^2$ on $\mathbb{A}^2$, the flat direction gives $H^\bullet = k[y] \otimes \Lambda(\xi_y) = \mathcal{O}(T^*[-1]\mathbb{A}^1_y)$. For $f = x^2y$, $H^0 = k[x,y]/(x^2, xy)$ is the non-lci ring of §1.1.5, and $H^{-1} = k[y] \cdot v$ with $v = x\xi_x - 2y\xi_y$, the Noether identity of the $\mathbb{G}_m$-symmetry $(x,y) \mapsto (tx, t^{-2}y)$. Adding a Koszul–Tate variable c^+ in degree -2 with $dc^+ = v$ kills H^{-1} but leaves $H^{-2} = k$ supported at the fixed point.

Planned content. [BV guide §5] $d(\xi_x\xi_y) = -x\,v$, so $H^{-1} = \mathcal{O}v/x\mathcal{O}v$; this is the $H_1 = k[y]$ of the naive Koszul complex in §1.1.5. After adding c^+: cycles in degree -2 are $\mathcal{O} \cdot z$, $z = \xi_x\xi_y + x\,c^+$, and $d(c^+\xi_x) = 2y\,z$, $d(c^+\xi_y) = x\,z$. Further $H^{-3} = 0$, $H^{-4} = k \cdot c^+z$, and so on, reflecting the stabilizer $\mathbb{G}_m$ at the origin. Exercise: $f = (x^2+y^2)^2$, with H^{-1} generated by the rotation. Leads to the derived quotient $[\mathrm{dCrit}\, f/\mathbb{G}_m]$ and to §1.2.3.

2 Algebraic topology

2.1 Pontryagin–Thom

2.1.1 Lie framing of stable normal bundle

Goal

Show that framings stabilize: for an n-dimensional CW complex M, $[M, \mathrm{O}(k)] \to [M, \mathrm{O}(n+k)]$ is surjective for $k \geq n+1$ and bijective for $k \geq n+2$, so a stable framing of the normal bundle is well defined.

Lemma 2.1. *For an n-dimensional CW complex M, the map $[M, \mathrm{O}(k)] \to [M, \mathrm{O}(n+k)]$ is surjective for $k \geq n+1$ and bijective for $k \geq n+2$.*

Proof. The fibration $\mathrm{O}(k) \to \mathrm{O}(n+k) \to V_n(\mathbb{R}^{n+k})$ has $(k-1)$-connected base, so $\pi_i(\mathrm{O}(k)) \to \pi_i(\mathrm{O}(n+k))$ is onto for $i \leq k-1$ and bijective for $i \leq k-2$. Obstruction theory over the n-skeleton of M gives the claim. □

2.1.2 Two stable framings of circle

Goal

Describe the two stable framings of S^1, the disk framing and the Lie framing, and show that the disk framing extends over the framed disk D^2, hence represents $0 \in \Omega_1^{fr}$.

A (stable) framing of S^1 is a trivialization of the normal bundle of the embedding

$$S^1 \hookrightarrow \mathbb{R}^2 \hookrightarrow \mathbb{R}^\infty = \mathbb{R}^2 \oplus \mathbb{R}e_3 \oplus \mathbb{R}e_4 \oplus \cdots \tag{75}$$

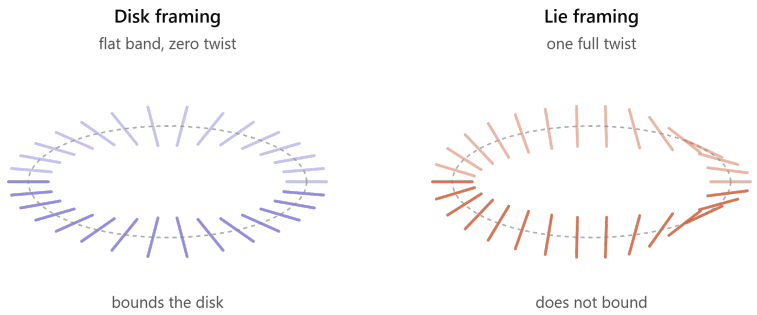
A trivial, or disk, framing is produced by adjoining parallel vectors $e_3, e_4, \cdots$ to the radial framing ν_r, where $TS^1 \oplus \nu_r = T\mathbb{R}^2$. That it defines the trivial element of Ω_1^{fr} can be seen from the framed cobordism

$$D^2 \hookrightarrow \mathbb{R}^3 \hookrightarrow \mathbb{R}^\infty = \mathbb{R}^3 \oplus \mathbb{R}e_4 \oplus \mathbb{R}e_5 \oplus \cdots \tag{76}$$

where D^2 is identified with the hemisphere, framed by the radial unit vectors in $\mathbb{R}^3$ and by adjoining $e_4, e_5, \cdots$ in $\mathbb{R}^\infty$.

$$D^2 \simeq \left\{ \boldsymbol{x} \in \mathbb{R}^3 \,\middle|\, |\boldsymbol{x}| = 1,\ x_3 \geq 0 \right\}, \qquad \boldsymbol{n}(\boldsymbol{x}) = \boldsymbol{x}/|\boldsymbol{x}| \tag{77}$$

Since $\pi_1(\mathrm{SO}) = \mathbb{Z}_2$, there is precisely one more framing of S^1, which we call Lie framing.



2.1.3 Hopf map and framing of circle

Goal

Show that the Lie-framed circle corresponds under Pontryagin–Thom, $\Omega_*^{fr} \simeq \pi_*(\mathbb{S})$, to the Hopf map η, and that $\eta \neq 0$ in $\pi_1(\mathbb{S}) = \mathbb{Z}/2$: $\mathbb{C}P^2$ is the mapping cone of η, and $\mathrm{Sq}^2 \neq 0$ on $\mathbb{C}P^2$ detects η stably (Proposition 2.2).

The Lie framing of a circle, defined in §2.1.2, maps to the nontrivial element of the first homotopy group of the sphere spectrum $\pi_1(\mathbb{S}) = \mathbb{Z}_2$. This element is induced from the Hopf map, which is a generator of $\pi_3(S^2) = \mathbb{Z}$. In this problem, we check that the Hopf map indeed corresponds to the Lie framing via the Pontryagin–Thom isomorphism

$$\Omega_*^{fr} \xrightarrow{\sim} \pi_*(\mathbb{S}). \tag{78}$$

We begin with the observation that $\mathbb{C}P^2$ is the mapping cone of the Hopf map. $\mathbb{C}P^2$ is a mapping cone due to the CW structure of $\mathbb{C}P^2 = \mathbb{C}P^1 \cup_\eta e^4$, where $\eta : S^3 \to S^2 \simeq \mathbb{C}P^1$ is an attaching map. Using the homogeneous coordinates on $\mathbb{C}P^2$, the 4-cell of $\mathbb{C}P^2$ can be represented as

$$\begin{array}{rccc} \Phi: & D^4 & \to & \mathbb{C}P^2 \\ & z & \mapsto & [z_1 : z_2 : \sqrt{1 - |z|^2}] \end{array} \tag{79}$$

so that $\Phi|_{\mathrm{int}\, D^4}$ is a homeomorphism onto $\mathbb{C}^2$, while $\eta = \Phi|_{S^3} : z \mapsto [z_1 : z_2]$ is the Hopf map. To see that the stable image of η is the generator of $\pi_1(\mathbb{S}) = \mathbb{Z}_2$, we prove the following principle:

Proposition 2.2. *For $f \in \pi_{n+k-1}(S^n)$ and $C_f = S^n \cup_f e^{n+k}$,*

$$\mathrm{Sq}^k \neq 0 \text{ on } \widetilde{H}^\bullet(\Sigma^j C_f; \mathbb{F}_2)\ \forall j \geq 0 \quad \Longrightarrow \quad f \neq 0 \text{ stably.} \tag{80}$$

Proof. Suppose $f = 0$ stably. Then for some j, $\Sigma^j C_f = S^{n+j} \cup_{\Sigma^j f} e^{n+j+k} \simeq S^{n+j} \vee S^{n+j+k}$. Since $\widetilde{H}^\bullet(S^{n+j} \vee S^{n+j+k}) \simeq \widetilde{H}^{n+j} \oplus \widetilde{H}^{n+j+k}$ as an abelian group, the only way in which Sq^k is possibly nontrivial is that it sends a generator of $\widetilde{H}^{n+j}$ to a nonzero element of $\widetilde{H}^{n+j+k}$. However, since the collapsing map

$$q : S^{n+j} \vee S^{n+j+k} \to S^{n+j}, \qquad q^* : \widetilde{H}^{n+j}(S^{n+j}) \to \widetilde{H}^{n+j}(S^{n+j} \vee S^{n+j+k}) \tag{81}$$

induces an isomorphism ($\because q \circ i = \mathrm{id}$ for an inclusion i), a generator $x \in \widetilde{H}^{n+j}(\Sigma^j C_f)$ can always be written as $x = q^* u$ for some generator u. Naturality of Sq^k then implies $\mathrm{Sq}^k(x) = q^*(\mathrm{Sq}^k(u)) = 0$. □

To apply Proposition 2.2 to $f = \eta$ and $n = k = 2$, we first see that $\mathrm{Sq}^2(x) = x^2 \neq 0$ on $\widetilde{H}^\bullet(\mathbb{C}P^2; \mathbb{F}_2) = \mathbb{F}_2[x]/(x^3)$, which is the $j = 0$ case of the hypothesis. Since the Steenrod squares commute with the suspension isomorphism, $\mathrm{Sq}^2 \neq 0$ for all $j \geq 1$ as well.

We have so far identified the generator of both sides of Eq. (78) (Lie framing and Hopf map). That the Pontryagin–Thom construction connects the two generators can be inferred from the fact that in the Hopf fibration, the preimages of two points on S^2 have a unit linking number. This is the case for the circle in §2.1.2 and its pushoff along the Lie framing.

2.1.4 Lie framings of tori and the Arf invariant

Goal (planned)

Use the multiplicativity of Pontryagin–Thom to show that the Lie-framed torus T^n represents η^n (T^1, T^2, T^3 do not bound, T^4 does), and detect $\eta^2 \neq 0$, which no single Steenrod square sees, by the Arf invariant of the Lie-framed T^2.

Planned content. [Guide §7, §8] The Pontryagin–Thom isomorphism Eq. (78) is a ring map, sending products of framed manifolds to smash products, so the Lie-framed torus T^n represents η^n: T^1, T^2, T^3 do not bound since $\eta^3 = 12\nu \neq 0$, while T^4 does since $\eta^4 = 0$ (Guide §7). For $\eta^2 \neq 0$, which no single Sq^k detects as in Proposition 2.2, define the quadratic refinement $q : H_1(\Sigma; \mathbb{F}_2) \to \mathbb{F}_2$ of a framed surface and compute $\mathrm{Arf}(q) = 1$ for the Lie-framed T^2, in contrast to the standard torus in $\mathbb{R}^3$ (Guide §8).

2.2 Steenrod squares and the Adams spectral sequence

Goal (planned)

Read Proposition 2.2 as the detection of η by h_1 in Adams filtration 1; use $\mathbb{H}P^2$ and $\mathbb{O}P^2$ for ν and σ, and the Adem relations for Hopf invariant one. Then compute $\mathrm{Ext}_{A(1)}(\mathbb{F}_2, \mathbb{F}_2)$ by hand and read off ko_*, hence $\Omega^{\mathrm{Spin}}_{\leq 7}$.

Planned content. [Guide §6, §10] Start by re-reading Proposition 2.2 as the statement that h_1 detects η in Adams filtration 1. Applying it to $\mathbb{H}P^2$ and $\mathbb{O}P^2$ (Sq^4, Sq^8) gives $\nu \neq 0$ and $\sigma \neq 0$; then Sq^n reads the Hopf invariant mod 2, the Adem relations make Sq^n decomposable unless $n = 2^j$, and Adams–Atiyah leave only $n = 2, 4, 8$ (Guide §6). Next, build the minimal resolution computing $\mathrm{Ext}^{s,t}_{A(1)}(\mathbb{F}_2, \mathbb{F}_2)$ for $s \leq 4$, draw the lightning-flash chart and read off $ko_* = \mathbb{Z}, \mathbb{Z}/2, \mathbb{Z}/2, 0, \mathbb{Z}, 0, 0, 0$; since $M\mathrm{Spin} \to ko$ is an isomorphism in degrees ≤ 7, this gives $\Omega^{\mathrm{Spin}}_{\leq 7}$ (Guide §10).

2.3 Image of J and the e-invariant

Goal (planned)

Define Adams' e-invariant from K-theory and the Chern character, compute $e(\eta) = \frac{1}{2}$ and $e(\nu) = \frac{1}{12}$ (real refinement $\frac{1}{24}$), and see how Bernoulli numbers give the image of J: the first appearance of height 1.

Planned content. [Guide §11] Define Adams' e-invariant $e_{\mathbb{C}} : \pi_{2k-1}\mathbb{S} \to \mathbb{Q}/\mathbb{Z}$ from the extension $0 \to \widetilde{K}(S^{2n+2k}) \to \widetilde{K}(C_f) \to \widetilde{K}(S^{2n}) \to 0$ and the Chern character, and compute $e_{\mathbb{C}}(\eta) = \frac{1}{2}$, $e_{\mathbb{C}}(\nu) = \frac{1}{12}$. The real refinement gives $e'_{\mathbb{R}}(\nu) = \frac{1}{24}$, which accounts for $\pi_3\mathbb{S} = \mathbb{Z}/24$; Bernoulli numbers then give the image of J in $\pi_{4k-1}\mathbb{S}$. This is the first appearance of height 1 (*KU*, §3.1).

2.4 Spin and Pin bordism

Goal (planned)

Compute three bordism groups that classify physical phases through explicit invariants: $\Omega_2^{\mathrm{Pin}^-} = \mathbb{Z}/8$ (Arf–Brown–Kervaire on $\mathbb{R}P^2$, the Klein bottle and T^2), $\Omega_4^{\mathrm{Spin}} = \mathbb{Z}$ (the K3 surface and Rokhlin's theorem), and $\Omega_4^{\mathrm{Pin}^+} = \mathbb{Z}/16$.

Planned content. [Guide §9] Three bordism groups that classify physical phases. $\Omega_2^{\mathrm{Pin}^-} = \mathbb{Z}/8$: $\mathbb{Z}/4$-valued quadratic enhancements and the Arf–Brown–Kervaire invariant, computed by hand on $\mathbb{R}P^2$, the Klein bottle and T^2 (the 8 of Fidkowski–Kitaev). $\Omega_4^{\mathrm{Spin}} = \mathbb{Z}$: the K3 surface with signature -16 and $\hat{A} = 2$, and Rokhlin's theorem. $\Omega_4^{\mathrm{Pin}^+} = \mathbb{Z}/16$. These feed the table of §4.2.

3 Chromatic homotopy

3.1 Complex orientations and formal group laws

Goal (planned)

Extract from a complex orientation the formal group law $e(L_1 \otimes L_2) = F(e(L_1), e(L_2))$; compute F and the $[p]$-series for $H\mathbb{Z}$ (height ∞) and KU (height 1), and build the Lazard ring by hand up to degree 4.

Planned content. [Guide §12] A complex orientation $x \in \widetilde{E}^2(\mathbb{C}P^\infty)$ gives the formal group law $F(x, y) = \mu^* x$, with $e(L_1 \otimes L_2) = F(e(L_1), e(L_2))$. Compute F and the $[p]$-series for $H\mathbb{Z}$ (additive, height ∞) and KU (multiplicative, height 1). Then build the Lazard ring by hand up to degree 4, checking the relation $2a_{22} = 2a_{11}a_{12} + 3a_{13}$ and that the coefficient of xy^3 is even.

3.2 Morava K-theory of $B\mathbb{Z}/p$

Goal (planned)

Compute $K(n)^*(B\mathbb{Z}/p) = K(n)^*[[x]]/(x^{p^n})$ by the Gysin sequence: height-n Morava K-theory sees exactly p^n dimensions of the infinite-dimensional $H^*(B\mathbb{Z}/p; \mathbb{F}_p)$. Compare with Hopkins–Kuhn–Ravenel.

Planned content. [Guide §13] For Morava K-theory with $K(n)^* = \mathbb{F}_p[v_n^{\pm 1}]$, $|v_n| = -2(p^n - 1)$ and $[p](x) = v_n x^{p^n}$, the Gysin sequence gives $K(n)^*(B\mathbb{Z}/p) = K(n)^*[[x]]/(x^{p^n})$, of rank p^n, although $H^*(B\mathbb{Z}/p; \mathbb{F}_p)$ is infinite. Künneth gives rank p^{nr} for $(\mathbb{Z}/p)^r$; compare with Hopkins–Kuhn–Ravenel, which counts commuting n-tuples.

4 Physics: SymTFT and invertible phases

4.1 Dijkgraaf–Witten, SymTFT and chromatic height

Goal (planned)

Prove $|\operatorname{Hom}(\mathbb{Z}^{n+1}, G)|/|G| = |\operatorname{Hom}(\mathbb{Z}^n, G)/G|$ and read it as $Z_{\mathrm{DW}}(T^{n+1}) = \dim \mathcal{H}_{\mathrm{DW}}(T^n)$. For a p-group, Hopkins–Kuhn–Ravenel identify this number with the $K(n)$-Euler characteristic of BG: height-n Morava K-theory counts the states of $(n{+}1)$d Dijkgraaf–Witten theory on T^n.

Planned content. [Guide §14] Prove the Burnside-type identity

$$\frac{|\operatorname{Hom}(\mathbb{Z}^{n+1}, G)|}{|G|} = \left|\operatorname{Hom}(\mathbb{Z}^n, G)/G\right|$$

and read the left side as $Z_{\mathrm{DW}}(T^{n+1})$ and the right side as $\dim \mathcal{H}_{\mathrm{DW}}(T^n)$ of $(n{+}1)$-dimensional Dijkgraaf–Witten theory. For a p-group G, Hopkins–Kuhn–Ravenel identify this number with

the $K(n)$-Euler characteristic of BG, which is $\operatorname{rk} K(n)^*(BG)$ when $K(n)^{\mathrm{odd}}(BG) = 0$: height-$n$ Morava K-theory counts the T^n Hilbert space of $(n{+}1)$d DW theory. Uses §1.2.3, §1.3.3 and §3.2. Note that "HKR" appears twice: Hochschild–Kostant–Rosenberg, $\mathcal{O}(LX) = \Omega^{-\bullet}_X$, and Hopkins–Kuhn–Ravenel, $E_n^*(BG)$ as functions on commuting n-tuples, i.e. on $L^n_p BG$.

4.2 Anderson duality and invertible phases

Goal (planned)

Use the Freed–Hopkins classification $[MTH, \Sigma^{n+1} I\mathbb{Z}]$ and its universal coefficient sequence to turn the bordism groups of §2.4 into the table of invertible phases in 0+1 to 3+1 dimensions: torsion gives SPT phases, the free part gives chiral theories.

Planned content. [Guide §15] Freed–Hopkins: deformation classes of reflection-positive invertible theories of symmetry type H are $[MTH, \Sigma^{n+1} I\mathbb{Z}]$. The universal coefficient sequence turns the bordism groups of §2.4 and §2.2 into the table of fermionic and bosonic invertible phases in 0+1 to 3+1 dimensions: torsion gives SPT phases, the free part gives chiral ($c \neq 0$) theories, and Rokhlin's 16 appears as the well-definedness of $p{+}ip$ (Steps 1–2). Step 3 connects the table to SymTFT, anomalies and chromatic height.

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